EPSR-test-ex95 Data file types nrtype 1, 3 and 4

Program(s): epsr | chained passes: 3 | verification: ALL IDENTICAL (all 3 passes)

What this case tests

Exercises the readers of the three non-GUDRUN data formats of the `epsr` engine of EPSR [1,2] ("read in data to be fitted" in `runepsr17`): `nrtype 1` two-column Q S(Q), `nrtype 3` GENIE histogram listing and `nrtype 4` GENIE point listing [3]. Users with data from other reduction programs depend on these paths.

Code exercised

Fortran (original, fortran_src/):

- `epsr_standalone_rev3.f`: `runepsr17` (data reading: list-directed `nrtype 1`, `(i7,3g14.7)` with header records for `nrtype 3/4`, histogram-to-centre Q conversion, ignored last point)

Rust (port, rust/epsr-core/src/):

- `datafile.rs`: `read_data_points` (fixed-format `(i7,3g14.7)` read with BLANK=NULL/PAD semantics, iostat chain of the header reads)
- `epsr.rs`: `Epsr::setup` (data sets, Q-point conversion)

Physical system

The EPSR-test system: a liquid mixture of 200 dimethyl sulphoxide (DMSO, (CH3)2SO) and 400 water molecules (DMSO mole fraction 1/3; 3200 atoms) at 300 K in a cubic box of side 33.12 Å (atomic number density 0.0881 atoms/Å^3). DMSO uses EPSR's Lennard-Jones + Coulomb reference potential with methyl rotational groups; water uses SPC/E Lennard-Jones and charge parameters [4] with EPSR's flexible intramolecular geometry (O-H 0.976 Å). The empirical potential is refined against three neutron diffraction data sets of deuterated DMSO in D2O, in null-scattering H2O/D2O and in H2O (ISIS SANDALS runs SLS03863, SLS03871, SLS03850), the isotope-substitution approach of Soper and Luzar [5]. Here the three data sets are supplied in three different formats: data 1 (`D6DMSO_D2O.nrt1`) as headerless list-directed `Q S(Q)` (errors set to 0) at bin centres; data 2 (`D6DMSO_HDO.nrt3`) in GENIE format with 4 header lines and histogram bin boundaries, converted to centres by EPSR; data 3 (`D6DMSO_H2O.nrt4`) in GENIE point format with an extra rubbish last point (S = 9.99) which EPSR ignores.

Inputs

`scripts/make_data.py` writes the three files from the base `.mint01` files (kept in the folder); `scripts/make_inp.sh` sets `datafile` and `nrtype` in the `.EPSR.inp` via `scripts/setkw.py`.

How the accuracy was verified

`commands.txt` (1 program call, repeated 3 times, each pass reading the files written by the previous one) was run with the Fortran reference build (`rust/tools/build_fortran_reference.sh`: the original sources, gfortran -O2 -fno-tree-vectorize, single thread) and with the Rust `epsr-rs` on identical copies of the inputs, and every file written was compared byte for byte (`rust/tools/run_ex_test.sh`, `compare_outputs.py`; only the date line of `.out` and the wall-clock columns of `fort.35` are masked). The Fortran output is stored in `fortran_reference.tar.gz`. Result of the final regression run of the complete port: every file identical in all 3 passes.
All three files give 594 points (log `Read 594 points from file ...`), the same as the original `.mint01` data; the identical outputs show that the record positioning of the Fortran was reproduced exactly.

Notes

Fortran quirks reproduced: for `nrtype 1` the first record is read and then `backspaced`, so the first line is data and the last point kept; for `nrtype 3/4` record 1 plus 4 records are read before backspacing, so record 5 is the first data line, and the last point is dropped.

References

[1] A. K. Soper, "Empirical potential Monte Carlo simulation of fluid structure", Chem. Phys. 202, 295-306 (1996).
[2] A. K. Soper, "Partial structure factors from disordered materials diffraction data: An approach using empirical potential structure refinement", Phys. Rev. B 72, 104204 (2005).
[3] A. K. Soper, "Empirical Potential Structure Refinement - EPSRshell: a user's guide", Technical Report RAL-TR-2011-012, STFC Rutherford Appleton Laboratory (2011).
[4] H. J. C. Berendsen, J. R. Grigera and T. P. Straatsma, "The missing term in effective pair potentials", J. Phys. Chem. 91, 6269-6271 (1987).
[5] A. K. Soper and A. Luzar, "A neutron diffraction study of dimethyl sulphoxide-water mixtures", J. Chem. Phys. 97, 1320-1331 (1992).
[6] GNU Fortran (gfortran) 15.2, GCC, https://gcc.gnu.org/fortran/ - reference build with -O2 -fno-tree-vectorize.